Band-1 Hadamard le64 Completeness Increment (H52 re-cert + full admissible coverage)

Finite construction/verification only (NOT Hadamard conjecture; 2004 OPEN; GATE-04 LOCKED)

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-14

Document class. Scientific computational / literature ledger increment (Band-1).
Program. Open-mathematics collapse list / Human+AI attack order / ZoraASI mission-control.
Parents. `2026-09-11_Band1_Computational_Experiment_hadamard` (lab `/workspace/experiments/band1/had`
`2026-09-11_Band1_Hadamard_single_order_probe` (lab `/workspace/experiments/band1/hadamard_order52/`)
Lab. `/workspace/experiments/band1/hadamard_le64_complete/` (mission-control mirror: `.../hadamard_le64_complete/`).
Cutoff. 2026-09-14.

Honesty stamp (non-negotiable).
This memo is a **finite construction / verification increment** only. It **does not** claim the full Hadamard conjecture. It **does not** construct or search order **2004**. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. `clay_closed = false`.

Abstract

We close the unique admissible gap left by the 2026-09-11 Band-1 Hadamard probe: order **52**. Re-certifying the Paley type-II matrix over GF(25) yields $\max|HH^\top - 52I| = 0$. Combining Sylvester, Paley I/II (prime fields), Kronecker products, and H52, we box-certify **all** admissible orders $n \in \{1, 2\} \cup \{4, 8, \ldots, 64\}$ (18/18). Headline: **coverage missing** $\leq 64 = [\ ]$. Kronecker extensions $H_{52} \otimes H_2$, H_4, H_8 certify orders **104, 208, 416**. Order **2004** and the full conjecture remain **OPEN**.

Keywords. Hadamard matrix; Paley construction; GF(25); Sylvester; Kronecker; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Classical necessary condition. $n \in \{1, 2\}$ or $4 \mid n$.

Finite lemma (this increment only). Explicit on-box constructions and exact int64 certificates for every admissible $n \leq 64$, plus three Kronecker extensions from H52. **Not** the Hadamard conjecture; **not** order 2004.

2. What changed vs 2026-09-11

Axis	Prior (2026-09-11)	This increment
Lab path	`hadamard/` + `hadamard_order52/`	`hadamard_le64_complete/`
Coverage ≤ 64	17/18; **missing = [52]**	**18/18; missing = []**
H52	single-order probe Paley II / GF(25)	**re-certified** + bits cross-check
Kronecker from H52	not run	**104, 208, 416** exact $HH^\top = nI$
Claim level	finite / lit	**same; conjecture OPEN; 2004 OPEN**

3. Methods

3.1 Sylvester

$H_1 = [1]$; doubling $H \mapsto \begin{pmatrix} H & H \\ H & -H \end{pmatrix}$ yields all powers of two.

3.2 Paley type I (prime $q \equiv 3 \pmod 4$)

Order $q+1$: border a Jacobsthal core $Q_{ij} = \chi(j-i)$ (Legendre) with a $+1$ row/column and $-I$ on the diagonal block.

3.3 Paley type II (prime $q \equiv 1 \pmod 4$)

Order $2(q+1)$ from the symmetric conference matrix $C = \begin{pmatrix} 0 & \mathbf{1}^\top \\ \mathbf{1} & Q \end{pmatrix}$: $H = \begin{pmatrix} C+I & C-I \\ C-I & -(C+I) \end{pmatrix}$.

3.4 Paley II over GF(25) $\rightarrow$ order 52

Field $\mathbb{F}_5[x]/(x^2+2)$ with $\alpha^2 \equiv 3 \pmod 5$ (irreducible: no root in $\mathbb{F}_5$). Quadratic character on the 25 elements; Jacobsthal Q is symmetric ($q \equiv 1 \pmod 4$); Paley II yields H_{52}. Reused / adapted from `hadamard_52_paley_II_gf25.py`.

3.5 Kronecker

If A, B are Hadamard of orders a, b then $A \otimes B$ is Hadamard of order ab. Used for 40 ($20 \otimes 2$), 56 ($28 \otimes 2$), and H52 extensions.

3.6 Certificate

Exact `int64` matmul; require $\max |HH^\top - nI| = 0$ and entries in $\{\pm 1\}$.

4. Unit tests

Test	Result
H_1 Sylvester	PASS
H_2 Sylvester (explicit $[[1,1],[1,-1]]$)	PASS
H_4 Sylvester	PASS
H_{12} Paley I ($q=11$)	PASS
Reject all-ones 4×4	PASS
H_{52} Paley II / GF(25), $\max \|HH^\top - 52I\| = 0$	PASS

All 6/6 unit tests PASS.

5. Headline: admissible coverage ≤ 64

n	Construction	$\max \|HH^\top - nI\|$	Wall (s)
1	Sylvester	0	$< 10^{-4}$
2	Sylvester	0	$< 10^{-4}$

n	Construction	$\max \|HH^\top - nI\|$	Wall (s)
4	Sylvester	0	$< 10^{-4}$
8	Sylvester	0	$< 10^{-4}$
12	Paley I $q = 11$	0	$< 10^{-4}$
16	Sylvester	0	$< 10^{-4}$
20	Paley I $q = 19$	0	$< 10^{-3}$
24	Paley I $q = 23$	0	$< 10^{-3}$
28	Paley II $q = 13$	0	$< 10^{-4}$
32	Sylvester	0	$< 10^{-4}$
36	Paley II $q = 17$	0	$< 10^{-3}$
40	Kronecker(H_{20}, H_2)	0	$< 10^{-3}$
44	Paley I $q = 43$	0	$< 10^{-3}$
48	Paley I $q = 47$	0	$< 10^{-3}$
52	**Paley II / GF(25)**	**0**	$< 10^{-3}$
56	Kronecker(H_{28}, H_2)	0	$< 10^{-3}$
60	Paley I $q = 59$	0	≈ 0.001
64	Sylvester	0	$< 10^{-3}$

Headline. Verified on box: **18 / 18**. Coverage missing $\leq 64 =$ [ ] (empty).

6. Kronecker extensions from H52

n	Construction	Verified	Wall (s)
104	$H_{52} \otimes H_2$	PASS	< 0.001
208	$H_{52} \otimes H_4$	PASS	≈ 0.007
416	$H_{52} \otimes H_8$	PASS	≈ 0.043

No OOM skips at these sizes on the 2026-09-14 box run.

7. Literature ledger refresh

7.1 Classical constructions

Sylvester (1867); Hadamard (1893) orders 12, 20; Paley (1933) I/II for prime powers; Williamson / Baumert-Golomb-Hall (1962) order 92; later large-order constructions (Kharaghani-Tayfeh-Rezaie 428; Djokovic 764; etc.).

7.2 August 2026 Alpoge et al. (reported)

Announcement that twelve historically open orders ≤ 2000 (668, 716, 892, 1132, 1244, 1388, 1436, 1676, 1772, 1916, 1948, 1964) now have explicit ± 1 matrices. **Not** re-verified matrix-by-matrix

in this increment (payloads large). Ledger treats ≤ 2000 as **reported-settled** pending archival citation.

7.3 Still OPEN

- **Order 2004** – smallest commonly cited remaining large gap; **NOT constructed / NOT searched** tonight.
- **Full Hadamard conjecture** – Band-3 **OPEN**.

8. What remains OPEN

1. Hadamard conjecture for all admissible $4k$ (Band-3).
2. Order **2004** (and larger unsettled orders).
3. GATE-04 Object 1 meter – collection **LOCKED** (orthogonal; no invented J).

9. Non-claims

- Not Clay-closed; not finished-ToE; GATE-04 LOCKED; `clay_closed = false`.
- Not RH / P≠NP / YM / Frankl / NS-solved / AC-solved.
- **Not** "Hadamard conjecture solved"; **not** order 2004; no Aug-2026 payload re-verification claimed.
- Finite construction / verification only.

10. Reproduction

```
cd /workspace/experiments/band1/hadamard_le64_complete
python3 verify_hadamard_le64.py
```

Wall-clock for the full suite on the 2026-09-14 box run: ≈ 0.06 **s**.

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `H52_pm1_bits.json`, `README.md`, this `PAPER.md`.

References (selected)

1. Paley, J. Math. Phys. 12 (1933) 311-320.
2. Wikipedia: Hadamard matrix; Paley construction (accessed 2026-09-14).
3. OEIS A007299.
4. Ion Nechita, "New Hadamard matrices" (13 Aug 2026 blog note on the announcement).
5. Prior Band-1 notes: `2026-09-11_Band1_Computational_Experiment_hadamard`; `2026-09-11_Band1_Hadam`